

DBMS — Lecture 3

Describing & Storing Data + DBMS Architecture

Learning Objectives

Understand how data is described in a DBMS.

Differentiate schema, instance, and metadata.

Understand basic DBMS architecture and components.

Describing Data

A DBMS stores values together with information describing their structure. Example: **STUDENT(RollNo, Name, Branch, Semester)**.

Schema, Instance & Metadata

Schema: The structure or design of a database; it changes relatively infrequently.

Instance: The actual data stored at a particular time; it changes frequently.

Metadata: Data about data, such as attribute name, data type, size, and constraints.

Basic Architecture

Users / Applications → DBMS → Database Storage

The DBMS acts as the interface between users/applications and stored data.

Major Components

Query Processor: Interprets and executes database queries.

Storage Manager: Handles storage and retrieval of data.

Transaction Management: Helps maintain correct database operations.

Authorization & Security: Controls access to data.

Database Storage: Contains data files, indexes, and related structures.

Example

Query: `SELECT * FROM Student;`

The DBMS processes the query and retrieves the required records from storage.

Key Takeaway

Schema describes structure, instance represents current data, and the DBMS manages access to stored data.